

ARITHMETIC AND GEOMETRIC PROGRESSIONS

1. Arithmetic Progression (AP)

Fundamental Theory

An AP is a sequence $a, a+d, a+2d, \dots$ where the difference between terms is constant. The n th term is $a_n = a + (n-1)d$

$$\text{Sum } (S_n) = (n/2) * [2a + (n-1)d]$$

Proof Sketch: Summing the AP forwards and backwards demonstrates that $2 * S_n = n * (2a + (n-1)d)$.

2. Geometric Progression (GP)

Fundamental Theory

A GP is a sequence $a, ar, ar^2, \dots$ with a common ratio r . The n th term is $a_n = a * r^{(n-1)}$

$$\text{Sum (S}_n\text{)} = a * (1 - r^n) / (1 - r)$$

Infinite Series: If $|r| < 1$, the sum converges to $S_\infty = a / (1 - r)$

3. Long-Answer Problem Bank

Problem: The Three-Term AP Challenge

The sum of three numbers in an AP is 27, and their product is 504. Find the numbers.

Detailed Solution:

1. Assume numbers as $a-d$, a , $a+d$.
2. Sum: $(a-d) + a + (a+d) = 27 \Rightarrow 3a = 27 \Rightarrow a = 9$.
3. Product: $(9-d) * 9 * (9+d) = 504$.
4. $(81 - d^2) = 504/9 = 56$.
5. $d^2 = 81 - 56 = 25 \Rightarrow d = \pm 5$.
6. If $d=5$, numbers are 4, 9, 14. If $d=-5$, numbers are 14, 9, 4.

Problem: The Three-Term GP Challenge

The sum of three numbers in a GP is $13/12$, and their product is -1 . Find the numbers.

Detailed Solution:

1. Assume numbers as a/r , a , ar .
2. Product: $(a/r) * a * (ar) = -1 \Rightarrow a^3 = -1 \Rightarrow a = -1$.
3. Sum: $-1/r - 1 - r = 13/12 \Rightarrow -1/r - r = 25/12$.
4. Multiply by $12r$: $-12 - 12r^2 = 25r \Rightarrow 12r^2 + 25r + 12 = 0$.
5. Solving quadratic: $r = -3/4$ or $-4/3$.
6. Resulting terms: $4/3$, -1 , $3/4$ (or vice-versa).